Suleyman Inaltekin

si20@rice.edu · sibuilds.github.io · github.com/sibuilds · LinkedIn

Statistics Ph.D. student at Rice University with experience in deep learning, computer vision, and large-scale predictive modeling. Builds machine learning models in PyTorch and NumPy, including neural networks and automatic differentiation from scratch. Research background in causal inference and probabilistic record linkage, with interests in probability theory, Bayesian inference, and physics-based efficient AI.

Education

Rice University Houston, TX
Ph.D. in Statistics | *Advisor: Prof. Meng Li* *Aug 2024 – Present* *(expected May 2030)*

- Transferred from the Political Science Ph.D. program (2024-26).
- Coursework: Measure-Theoretic Probability, Deep Learning, Causal Inference, Statistical Foundations of Generative AI

University of Vienna Vienna, Austria
B.A. Political Science, minor in Mathematics | *GPA: 3.7/4.0* *2019 – 2023*

Memorial International School of Tirana Tirana, Albania
Cambridge International A Levels: Mathematics (A), History (A), Psychology (A)* *2019*

Research Experience

Graduate Researcher, Department of Statistics, Rice University 2024 – Present
Project title, e.g., Langevin and Oscillator-Based Samplers for Energy-Based Models *Advisor: Prof. Meng Li*

- Developing [method] for [problem]; [result, e.g., 5× faster than MCMC on 1M rows at equal accuracy].
- Released [method] as an open-source R/Python package; [presented at JSM 2026 / paper in prep].

Projects

PunchTracker: an AI Boxing Coach from a Phone Video | code Python, MediaPipe, OpenCV, SciPy

- Built a computer-vision pipeline that tracks 33 body keypoints per frame and detects, counts, and classifies punches (jab, cross, hook, uppercut) from a single smartphone video, with no wearable sensors.
- Turned wrist-extension trajectories into a time series and detected punches with velocity-thresholded peak finding; [X]% count accuracy against hand-labeled sparring rounds.

Swipe or Stay: Predicting Attention Retention in Short Videos | code PyTorch, pandas

- Trained a neural embedding model on 4.7M user–video interactions (KuaiRec) to predict watch time, a core engagement signal in short-video feed ranking (TikTok, Reels, Spotlight).
- Diagnosed and corrected duration bias (longer videos mechanically get lower watch ratios) with duration-conditional labels; [X]% lower error than matrix-factorization and popularity baselines.

Autograd & Neural Networks from Scratch | code Python, NumPy

- Built a micrograd-style autograd engine and MLP in pure NumPy; verified gradients against finite differences and reached [X]% test accuracy on handwritten digits.

Teaching

Teaching Assistant, Rice University | STAT 310: Probability & Statistics Fall 2026 – Present

Skills

Programming: Python (PyTorch, NumPy, pandas, scikit-learn, OpenCV), R (tidyverse, Stan), SQL, Git, LaTeX
Methods: causal inference (DiD, IV, RDD, matching), A/B testing & experimental design, Bayesian modeling, MCMC & Langevin sampling, energy-based models, deep learning, computer vision
Languages: Turkish (native), Albanian (native), English (fluent), German (fluent)
Interests: swimming, chess, card magic, neuromorphic computing